Panskura Banamali College (Autonomous)

B.Sc.(H) 4th Semester Examination - 2022 (Online Mode)

Sub: BCA **Paper: GE4T**

Paper Name: Graph Theory

F.M.: 60 **Time : 2 Hours**

Answer any four questions: **4x15=60**

1.a) Can a simple graph have 5 vertices and 12 edges? If so, draw it; if not, explain why it is not possible to have such a graph.

b) How many more edges are there in the complete graph K7 than in the complete graph K5?

c) Explain BFS with example. 3+2+10=15

2. Explain Dijkstra algorithm for shortest path problem. Write Kuratowski's theorem . 10+5=15

3. Define binary tree .Write down its properties . Prove that every tree has either one or two centers. 4+6+5=15

4. Describe minimum spanning tree. Explain Prim's and Kruskal algorithm to find minimum sanning tree. 5+10=15

5.Explain DFS with example. For any positive integer n, if G is a connected graph with n vertices and n-1 edges, then prove that G is a tree . 10+5=15

6. Draw the following graphs:

(i) the null graph N5;

(ii) the complete graph K6;

(iii) the complete bipartite graph K7 4;

(iv) the union of Kl 3 and W4;

(v) the complement of the cycle graph C4. 10